EMPOWER-HER: A PRIVACY PRESERVING AI FRAMEWORK FOR WOMEN ENTREPRENEURSHIP

Project Stage – 1 Report (CS753PC)

Submitted

in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

By

B. PRANITHA (23261A0576)

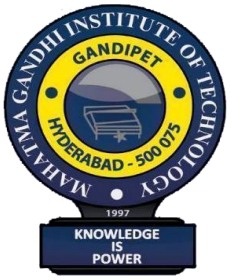
BOLLU BHAVYA SRI (23261A0577)

C. MANI CHANDRA (23261A0578)

Under the guidance of

Mrs. K. Sunitha

(Assistant Professor)



Department Of Computer Science and Engineering,

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY(A),

GANDIPET, HYDERABAD-500 075, INDIA.

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MAHATMA GANDHI INSTITUTE OF TECHNOLOGY

(Affiliated to Jawaharlal Nehru Technological University Hyderabad) Gandipet,
Hyderabad-500 075, Telangana (India)

CERTIFICATE

This is to certify that the thesis entitled **"EMPOWER-HER:A PRIVACY PRESERVING AI FRAMEWORK FOR WOMEN ENTREPRENEURSHIP "** is being submitted by **Ms. B. Pranitha** bearing Roll No: **23261A0576**, **Ms. Bollu Bhavya Sri** bearing Roll No: **23261A0577** and **Mr. C. Mani Chandra** bearing Roll No: **23261A0578** in partial fulfillment for award of **Bachelor of Technology** in **Computer Science and Engineering** to Mahatma Gandhi Institute of Technology(A) is a record of bona-fide work carried out by them under our guidance and supervision. The results embodied in this project have not been submitted to any other University or Institute for the award of any degree or diploma.

Project Guide
Mrs. K. Sunitha
Asst. Professor, Dept. of CSE

Head of the Department
Dr. T.V. Rajinikanth
Professor, Dept. of CSE

DECLARATION

This is to certify that the work described in this report entitled "**EMPOWER-HER: A PRIVACY PRESERVING AI FRAMEWORK FOR WOMEN ENTREPRENEURSHIP**" is a record of work done by us in the Department of Computer Science and Engineering, Mahatma Gandhi Institute of Technology, Hyderabad. No part of the work is copied from books/journals/internet and wherever the portion is taken, the same has been duly referred in the text. The report is based on the work done entirely by us and not copied from any other source.

B. PRANITHA
(23261A0576)

BOLLU BHAVYA SRI
(23261A0577)

C. MANI CHANDRA
(23261A0578)

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B. PRANITHA
(23261A0576)

BOLLU BHAVYA SRI
(23261A0577)

C. MANI CHANDRA
(23261A0578)

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
CSS	Cascading Style Sheets
DRF	Django REST Framework
HE	Homomorphic Encryption
HTML	HyperText Markup Language
LLM	Large Language Model
ML	Machine Learning
NLP	Natural Language Processing
OCR	Optical Character Recognition
REST	Representational State Transfer
ZKP	Zero-Knowledge Proof

ABSTRACT

Women entrepreneurs often face challenges in accessing government schemes, securing funding, validating business ideas, and sharing sensitive information with mentors, investors, and financial institutions. Most of the digital platforms provide information but lack in intelligent guidance and strong privacy protection, making it difficult to safeguard confidential business ideas and personal data.

Empower-Her is a privacy-preserving AI framework designed to address these challenges. The framework uses Homomorphic Encryption to securely process sensitive documents, such as identity and financial records, without exposing the original data. It also employs Zero-Knowledge Proofs (ZKP), enabling entrepreneurs to demonstrate the authenticity and progress of their business ideas without revealing confidential details.

In addition to secure data sharing, Empower-Her offers AI-powered business idea validation, personalized government scheme recommendations, local market demand analysis, multilingual voice assistance, OCR-based document verification, loan EMI estimation, and women-centric safety insights. By integrating privacy-preserving technologies with intelligent decision support, Empower-Her creates a secure and trustworthy ecosystem that helps women entrepreneurs confidently protect their innovations, access financial opportunities, and make informed business decisions, creating a supportive space where women can grow their businesses with confidence and security.

Keywords: Women Entrepreneurship, Artificial Intelligence (AI), Homomorphic Encryption, Zero-Knowledge Proofs (ZKP), Business Idea Validation, Government Scheme Recommendation, Market Demand Analysis, Multilingual Voice Assistance, Optical Character Recognition (OCR), Loan EMI Estimation, Privacy-Preserving Technologies, Document Verification, Women's Safety.

1. INTRODUCTION

Women entrepreneurs play an important role in economic development by creating employment opportunities, encouraging innovation, and contributing to society. However, many women who want to start or develop a business, face challenges in accessing government schemes, finding suitable funding opportunities, validating their business ideas, and receiving proper guidance. They may also find it difficult to identify potential customers, understand market competition, and make informed business decisions.

With the advancement of Artificial Intelligence (AI), web technologies, and privacy-preserving techniques, new possibilities have emerged to support women entrepreneurs through digital platforms. AI can help users evaluate their business ideas, understand market opportunities, and receive personalized suggestions. At the same time, secure data management and privacy technologies can help protect sensitive business information from unauthorized access.

This project focuses on developing Empower-Her, an AI-based platform designed to support women entrepreneurs by providing business idea validation, information about relevant government schemes, funding opportunities, and guidance for business development. The system also incorporates privacy protection mechanisms, including Zero-Knowledge Proof (ZKP)-based verification, to support the secure sharing of sensitive information. By combining intelligent assistance with privacy-focused features, Empower-Her aims to create a more accessible, secure, and supportive environment for women to develop their entrepreneurial ideas.

1.1 Problem Definition

Women entrepreneurs often face several difficulties when starting or expanding their businesses. Finding reliable information about government schemes, financial assistance, and funding opportunities can be challenging, as the information is spread across different websites and platforms. Many aspiring entrepreneurs also lack access to expert guidance to evaluate whether their business ideas are practical, identify their target customers, and understand existing competition.

Existing digital platforms may provide business-related information, but users often need to search through multiple sources to find information relevant to their specific needs. General-purpose AI chatbots can provide suggestions, but their responses may not always be based on reliable or up-to-date market information. This makes it difficult

for users to make informed decisions about their business ideas.

Another important challenge is the privacy of sensitive business information. Entrepreneurs may need to share business plans, financial details, and personal information with mentors, investors, or other stakeholders. Sharing such information without appropriate privacy protection may create concerns about unauthorized access or misuse. Conventional systems that rely only on authentication and access control may not provide the level of privacy required for certain verification processes.

Therefore, the main problem is to develop an integrated platform that provides intelligent business guidance, helps women entrepreneurs discover relevant opportunities, and protects confidential information. Empower-Her aims to address these challenges by combining AI-based business idea validation, government scheme and funding information, and privacy-preserving verification mechanisms in a single, user-friendly application.

1.2 Existing Applications

Several existing digital platforms provide services related to entrepreneurship, government schemes, funding, and business development. Government portals offer information about business registration, financial assistance, and schemes for entrepreneurs. Platforms such as Startup India provide resources and support for startups, while other financial and business websites help users explore funding opportunities and business-related information.

AI-based chatbots and business advisory tools can also assist users by generating business ideas, suggesting improvements, and answering general business-related questions. These tools make information more accessible, but their recommendations may require additional verification and may not always be tailored to the specific requirements of women entrepreneurs.

Although these applications provide useful services, their functionalities are often distributed across separate platforms. Users may need to switch between multiple websites to validate an idea, identify suitable schemes, explore funding opportunities, and obtain business guidance. Furthermore, many platforms focus mainly on information delivery and do not integrate business idea validation with privacy-preserving verification.

As a result, there is a need for a unified system that combines intelligent business assistance, access to relevant entrepreneurial resources, and secure handling of confidential information. Empower-Her is proposed to address these limitations by bringing these functionalities together in one platform.

1.3 Proposed Application

The proposed system, Empower-Her, is an AI-based web application designed to help women entrepreneurs explore, evaluate, and develop their business ideas. It provides a centralized platform through which users can access business guidance, discover relevant government schemes and funding opportunities, and receive suggestions for improving their business plans.

The system allows users to enter details such as their business name, idea description, business category, target customers, and location. Based on the provided information, the AI-based business idea validation module generates feasibility feedback, identifies potential customer groups, highlights possible competition, and suggests improvements. The platform also helps users discover relevant government schemes and funding resources using available information from reliable sources. To improve information security, Empower-Her incorporates privacy-preserving mechanisms, including Zero-Knowledge Proof (ZKP)-based verification. This approach can allow a user to demonstrate that a particular condition is satisfied without revealing the underlying sensitive information, where the verification requirements support such proofs. Authentication and access control are also used to restrict access to protected resources.

By integrating AI-based business guidance, entrepreneurial resource discovery, and privacy-focused verification, Empower-Her aims to reduce the effort required to find relevant information and support women entrepreneurs in making informed business decisions. The system provides a common digital platform that combines business assistance with the protection of confidential information.

1.4 Objectives of Empower-Her

- To develop a user-friendly web application that supports women entrepreneurs in exploring and developing their business ideas.
- To provide AI-based business idea validation, including feasibility feedback, potential customer identification, competition insights, and improvement suggestions.
- To help users discover relevant government schemes, financial assistance programs, and funding opportunities using reliable information sources.
- To provide personalized business guidance based on the information and requirements entered by users.

1.5 Requirements Specification

Software Requirements

- Operating System: Windows 10/11, Linux, or macOS
- Programming Language: Python 3.9 or later
- Framework: Django(python)
- Development Environment: VS Code/PyCharm
- Browser: Chrome/Firefox/Edge (latest versions)

Hardware Requirements

1. Development System

- CPU: Multicore processor (Intel i3 or later)
- RAM: 8 GB minimum
- Storage: 20 GB free disk space

2. Deployment/Testing System

- Local network or virtual environment
- Internet connectivity for updates and dependencies

2. LITERATURE SURVEY

[1] "Privacy-Preserving Attribute-Based Access Control Using Homomorphic Encryption" by M. Kerl, U. Bodin, and O. Schlén (2025)

This paper presents a privacy-preserving attribute-based access control system using Paillier Homomorphic Encryption (HE). The methodology focuses on performing attribute-based access control while protecting sensitive user attributes through encrypted policy evaluation. The system ensures privacy and supports controlled access to sensitive information. However, encrypted policy evaluation introduces additional computational overhead. The research highlights the need for integrating AI-based business guidance and recommendation systems with privacy-preserving access control mechanisms.

[2] "Mathematical Proposal for Securing Split Learning Using Homomorphic Encryption and Zero-Knowledge Proofs" by A. Kokaj and E. Mollakuqe (2025)

This research proposes a mathematical framework for securing split learning using Homomorphic Encryption (HE), Zero-Knowledge Proofs (ZKPs), and CKKS/BFV encryption schemes. The methodology combines encryption techniques and anomaly detection to improve the security and privacy of split learning. It protects privacy, maintains model integrity, and supports attack detection. However, cryptographic operations increase computational complexity. The research gap lies in integrating these privacy-preserving techniques into a women-entrepreneurship platform to provide secure and reliable business support.

[3] "Approximate Homomorphic Encryption Based Privacy-Preserving Machine Learning: A Survey" by J. Yuan, W. Liu, J. Shi, and Q. Li (2025)

This paper presents a survey of approximate Homomorphic Encryption techniques used in privacy-preserving machine learning. The methodology involves reviewing and analyzing various HE techniques for secure machine learning applications. The study summarizes existing techniques and identifies important research gaps in privacy-preserving machine learning. However, the survey-based approach does not provide an end-to-end practical implementation. The research gap is the need for practical integration of privacy-preserving machine learning techniques into personalized entrepreneurship services.

[4] "Secure Federated Learning via Neural Cryptography with Homomorphic Operations" by E. Sele, F. O. Catak, J. Seo, and M. Kuzlu (2025)

This research proposes a secure federated learning approach using neural cryptography

and Homomorphic Encryption operations. The methodology focuses on performing secure federated aggregation while protecting model updates and maintaining data privacy. The system supports secure collaborative model training without directly sharing sensitive data. However, the approach introduces architectural complexity and provides limited cryptographic guarantees. The research gap is the lack of specific protection for women entrepreneurs' business and financial information in privacy-preserving business support platforms.

[5] "Privacy-Preserving Machine Learning: ANN Activation Function Estimators for Homomorphic Encrypted Inference" by M. R. Abou Harb and B. Celiktas (2025)

This paper presents a privacy-preserving machine learning approach using Artificial Neural Network (ANN) activation function estimators for Homomorphic Encryption-based inference. The methodology develops estimators for activation functions such as Sigmoid and Tanh to enable efficient inference over encrypted inputs. The approach supports more efficient machine learning inference while protecting sensitive input data. However, encrypted inference can remain computationally expensive. The research gap is the lack of integration between privacy-preserving inference and entrepreneurship guidance or scheme recommendation systems.

[6] "Practical Solutions in Fully Homomorphic Encryption: A Survey Analyzing Existing Acceleration Methods" by Y. Gong et al. (2024)

This paper presents a survey of practical solutions for improving the performance of Fully Homomorphic Encryption (FHE). The methodology reviews algorithmic, hardware-based, and system-level acceleration methods for encrypted computation. The study identifies existing techniques that can improve computational efficiency and reduce the performance overhead of FHE. However, the effectiveness of acceleration methods depends on the specific workload and implementation. The research gap is the need for application-specific privacy protection and efficient encrypted computation within an accessible business support platform.

[7] "Advanced Privacy-Preserving Framework Using Homomorphic Encryption and Adaptive Privacy Parameters for Scalable Big Data Analysis" by R. Shanthi et al. (2024)

This research proposes an advanced privacy-preserving framework using Homomorphic Encryption and adaptive privacy parameters for scalable big data analysis. The methodology combines HE techniques with adaptive privacy parameter selection to enable secure analysis of large-scale datasets. The framework provides scalable privacy

protection and supports secure big data processing. However, the approach involves high computational requirements. The research gap is the need to integrate scalable privacy-preserving techniques with personalized entrepreneurial services to provide secure and efficient business recommendations.

[8] "UniHENN: Designing Faster and More Versatile Homomorphic Encryption-Based CNNs Without im2col" by H. Choi, J. Kim, S. Kim, S. Park, J. Park, W. Choi, and H. Kim (2024)

This paper proposes UniHENN, a Homomorphic Encryption-based Convolutional Neural Network (CNN) architecture designed to improve computational efficiency without using the im2col operation. The methodology focuses on designing HE-based CNNs that reduce computational overhead and improve memory utilization. The approach improves processing speed and memory efficiency for encrypted machine learning applications. However, Homomorphic Encryption-based models remain resource-intensive. The research gap is the lack of application-specific adaptation for protecting confidential business data and providing entrepreneurship-focused recommendations.

[9] "Privacy-Preserving Embedding via Look-Up Table Evaluation with Fully Homomorphic Encryption" by J.-Y. Kim, S. Park, J. Lee, and J. H. Cheon (2024)

This research proposes a privacy-preserving embedding approach using Look-Up Table (LUT) evaluation with Fully Homomorphic Encryption (FHE). The methodology uses FHE and look-up tables to perform embedding computations while protecting sensitive input information. The approach protects input privacy during computation and enables secure embedding evaluation. However, the method involves high computational costs due to Fully Homomorphic Encryption. The research gap is the need for practical adaptation of privacy-preserving embedding techniques into business-support applications that provide secure and personalized recommendations.

[10] "FedNIC: Enhancing Privacy-Preserving Federated Learning via Homomorphic Encryption Offload on SmartNIC" by S. Choi, D. Patel, D. Zad Tootaghaj, L. Cao, F. Ahmed, and P. Sharma (2024)

This paper presents FedNIC, a privacy-preserving federated learning framework that offloads Homomorphic Encryption operations to Smart Network Interface Cards (SmartNICs). The methodology focuses on performing encrypted operations through specialized network hardware to improve federated learning performance. The approach reduces CPU overhead and improves computational performance during secure federated learning. However, the framework requires specialized hardware, which may increase deployment complexity and cost. The research gap is the lack of an accessible,

integrated privacy-preserving platform specifically designed to support women entrepreneurs through secure business guidance and recommendations.

[11] "Key Factors Underlying Innovation in the Context of Female Entrepreneurship: Opportunity Perceptions, Digitalisation, and Economic Policy" by M. S. Castaño Martínez and S. G. Romero Medina (2025)

This paper examines the key factors influencing innovation in female entrepreneurship, focusing on opportunity perceptions, digitalisation, and economic policy. The methodology involves analyzing the relationship between entrepreneurial opportunities, digital technologies, and economic policies to understand innovation among women entrepreneurs. The study identifies important factors that contribute to innovation and provides insights into female entrepreneurship. However, the research does not implement an AI-based recommendation platform. The research gap is the need for personalized AI guidance that connects women entrepreneurs with suitable government schemes, funding opportunities, and business support services.

[12] "Determinants Influencing the Entrepreneurial Success of MSMEs in Emerging Economies: A Study of Indian Women Entrepreneurs" by M. Nayak, P. M. Nayak, and H. G. Joshi (2025)

This research investigates the factors influencing the entrepreneurial success of Micro, Small, and Medium Enterprises (MSMEs) led by women entrepreneurs in India. The methodology focuses on studying the factors affecting entrepreneurial success in emerging economies, particularly in the Indian context. The study provides context-specific insights into the factors contributing to the success of women-led businesses. However, the research involves questionnaire and sampling limitations and does not provide automated recommendations. The research gap is the need for an AI-based guidance system that provides personalized recommendations according to the business context, requirements, and challenges faced by women entrepreneurs.

[13] "Challenges and Future of Pradhan Mantri Mudra Yojana (PMMY) in Promoting Women Entrepreneurship in India" by M. Dhanalakshmi and E. Karthikeyan (2025)

This paper examines the challenges and future opportunities of the Pradhan Mantri Mudra Yojana (PMMY) in promoting women entrepreneurship in India. The methodology focuses on analyzing PMMY, financial inclusion, and the barriers faced by women entrepreneurs in accessing financial assistance. The study highlights funding opportunities and identifies important barriers affecting women entrepreneurs. However, the research focuses on a single government scheme and does not provide a technical

implementation. The research gap is the need for an intelligent recommendation system that connects women entrepreneurs with suitable government schemes, financial assistance, and funding opportunities based on their eligibility.

[14] "The Role of Digitalization and Education in Empowering Egyptian Women Entrepreneurs" by D. Salman, N. Fawzy, and Z. Zaazou (2025)

This research investigates the role of digitalization and education in empowering women entrepreneurs in Egypt. The methodology uses surveys and Partial Least Squares Structural Equation Modeling (PLS-SEM) to study the relationship between digitalization, education, and women's entrepreneurial empowerment. The study provides empirical findings regarding the contribution of digital technologies and education to entrepreneurship. However, the research is focused on Egypt, which limits its generalization to other regions. The research gap is the need for an accessible AI-assisted guidance platform that provides personalized business support and suitable government scheme recommendations for both rural and urban women entrepreneurs.

[15] "Enhancing Women Entrepreneurship Through Resource Orchestration and Digital Technologies" by P. Prabha, A. Swati, and R. Kumar (2025)

This paper explores the role of resource orchestration and digital technologies in enhancing women entrepreneurship. The methodology uses Structural Equation Modeling (SEM) to study the relationship between digital technologies, resource orchestration, and women's entrepreneurial growth. The study establishes a connection between digitalization and business development and highlights the importance of effectively managing resources. However, the research lacks privacy and security mechanisms for protecting sensitive entrepreneurial information. The research gap is the need for integrating secure, personalized, and privacy-preserving entrepreneurial support into a digital platform that provides business guidance and recommendations.

[16] "Transforming Rural Women's Lives in India: The Impact of Microfinance and Entrepreneurship on Empowerment in Self-Help Groups" by A. Pandhare, P. N. Bellampalli, and N. Yadava (2024)

This research examines the impact of microfinance and entrepreneurship on the empowerment of rural women in India through Self-Help Groups (SHGs). The methodology focuses on studying the role of microfinance and entrepreneurial activities in improving the financial independence and empowerment of rural women. The study highlights the importance of financial inclusion and entrepreneurship in improving women's economic conditions. However, the research does not focus on digital privacy or security mechanisms. The research gap is the need for an accessible digital guidance

platform that provides rural women entrepreneurs with personalized business recommendations while ensuring the privacy and security of their sensitive information.

[17] "Secure Processing and Distribution of Data Managed on Private InterPlanetary File System Using Zero-Knowledge Proofs" by K. Shibano, K. Ito, H. Han, T. T. Chu, W. Ozaki, and G. Mogi (2024)

This paper proposes a secure approach for processing and distributing data managed on a private InterPlanetary File System (IPFS) using Zero-Knowledge Proofs (ZKPs). The methodology uses private IPFS and zk-SNARKs to enable secure data verification while protecting sensitive information. The approach supports data integrity and privacy during secure data processing and distribution. However, the system has limited large-scale evaluation. The research gap is the need to adapt these secure data verification techniques for confidential document verification and privacy-preserving information management in an entrepreneurship support platform.

[18] "ZKML: An Optimizing System for ML Inference in Zero-Knowledge Proofs" by B.-J. Chen, S. Waiwitlikhit, I. Stoica, and D. Kang (2024)

This research presents ZKML, an optimization system designed to improve machine learning inference using Zero-Knowledge Proofs. The methodology focuses on optimizing machine learning inference representations and proof generation using ZKPs. The system reduces the computational cost associated with generating zero-knowledge proofs and improves the efficiency of privacy-preserving machine learning applications. However, Zero-Knowledge Proof generation remains computationally expensive. The research gap is the need for practical implementation of secure AI-based recommendations for women entrepreneurs while reducing computational overhead and protecting sensitive business information.

[19] "Privacy-Preserving Machine Learning (PPML) Inference for Clinically Actionable Models" by B. Balaban et al. (2025)

This paper presents a privacy-preserving machine learning inference approach for clinically actionable models. The methodology applies privacy-preserving techniques to machine learning inference in healthcare applications to protect sensitive input data during computation. The approach ensures the confidentiality of sensitive information and supports secure machine learning inference. However, the privacy-preserving techniques introduce increased latency and computational overhead. The research gap is the need to extend privacy-preserving inference techniques beyond healthcare and integrate them into entrepreneurship support platforms to provide secure and personalized business recommendations.

[20] "Homomorphic Encryption for Machine Learning Applications with CKKS Algorithms: A Survey of Developments and Applications" by L. Wu, X. A. Wang, J. Liu, Y. Su, Z. Tu, W. Liu, H. Lei, D. Tang, Y. Cao, and J. Zhang (2025)

This paper presents a comprehensive survey of Homomorphic Encryption techniques based on the CKKS algorithm for machine learning applications. The methodology involves reviewing existing developments and applications of CKKS-based Homomorphic Encryption in privacy-preserving machine learning. The study provides a detailed overview of CKKS applications and highlights the potential of encrypted computation for protecting sensitive information during machine learning operations. However, the survey-based approach does not provide a complete practical implementation. The research gap is the need for application-level integration of CKKS-based Homomorphic Encryption into personalized and privacy-preserving services that provide secure business guidance and recommendations for women entrepreneurs.

Table 2.1 : Literature survey Table

S. No.	Authors	Title	Year	Journal / Source	Methodology / Description	Merits	Demerits
1	M. Kerl, U. Bodin, O. Schelén	Privacy-Preserving Attribute-Based Access Control Using Homomorphic Encryption	2025	Cybersecurity (Springer)	Uses Paillier HE for privacy-preserving attribute-based access control.	Protects sensitive attributes and supports controlled access.	Encrypted policy evaluation adds computation overhead.
2	A. Kokaj and E. Mollakuqe	Mathematical Proposal for Securing Split Learning Using Homomorphic Encryption and Zero-Knowledge Proofs	2025	Applied Sciences	Combines HE, ZKPs, CKKS/BFV and anomaly detection to secure split learning.	Protects privacy and model integrity; supports attack detection.	Cryptographic operations increase computational complexity.
3	J. Yuan, W. Liu, J. Shi, Q. Li	Approximate Homomorphic Encryption Based Privacy-Preserving Machine Learning: A Survey	2025	Artificial Intelligence Review	Surveys approximate HE techniques for privacy-preserving machine learning.	Summarizes techniques and identifies research gaps.	Survey-based; no end-to-end implementation.
4	E. Sele, F. O. Catak, J. Seo, M. Kuzlu	Secure Federated Learning via Neural Cryptography with Homomorphic Operations	2025	Discover Artificial Intelligence	Uses neural cryptography and homomorphic operations for secure federated aggregation.	Supports secure collaborative model training.	Adds architectural complexity and has limited cryptographic guarantees.
5	M. R. Abou Harb and B. Celiktas	Privacy-Preserving Machine Learning: ANN Activation Function Estimators for Homomorphic Encrypted Inference	2025	IEEE Access	Develops ANN estimators for HE-based Sigmoid and Tanh inference.	Supports more efficient ML inference on encrypted inputs.	Encrypted inference can remain computationally expensive.
6	Y. Gong et al.	Practical Solutions in Fully Homomorphic Encryption: A Survey Analyzing Existing Acceleration Methods	2024	Cybersecurity	Reviews algorithmic, hardware and system-level acceleration methods for FHE.	Identifies methods that may improve encrypted-computation efficiency.	Performance depends on the method and workload.
7	R. Shanthi et al.	Advanced Privacy-Preserving Framework Using Homomorphic Encryption and Adaptive Privacy Parameters for Scalable Big Data Analysis	2024	International Journal of Intelligent Systems and Applications in Engineering	Uses HE and adaptive privacy parameters for secure big-data analysis.	Provides scalable privacy protection.	High computational requirements.

8	H. Choi, J. Kim, S. Kim, S. Park, J. Park, W. Choi, H. Kim	UniHENN: Designing Faster and More Versatile Homomorphic Encryption-Based CNNs Without im2col	2024	IEEE Access	Designs HE-based CNNs without the im2col operation.	Improves speed and memory efficiency.	HE-based models remain resource-intensive.
9	J.-Y. Kim, S. Park, J. Lee, J. H. Cheon	Privacy-Preserving Embedding via Look-Up Table Evaluation with Fully Homomorphic Encryption	2024	Proceedings of ICML	Uses FHE and look-up tables for privacy-preserving embeddings.	Protects input privacy during computation.	High FHE computation cost.
10	S. Choi, D. Patel, D. Zad Tootaghaj, L. Cao, F. Ahmed, P. Sharma	FedNIC: Enhancing Privacy-Preserving Federated Learning via Homomorphic Encryption Offload on SmartNIC	2024	Frontiers in Computer Science	Offloads HE operations to SmartNIC for secure federated learning.	Reduces CPU overhead and improves performance.	Requires specialized hardware.
11	M. S. Castaño Martínez and S. G. Romero Medina	Key Factors Underlying Innovation in the Context of Female Entrepreneurship: Opportunity Perceptions, Digitalisation, and Economic Policy	2025	Journal of International Entrepreneurship	Examines opportunity perception, digitalisation and economic policy in female entrepreneurship.	Identifies factors relevant to innovation.	Does not implement an AI recommendation platform.
12	M. Nayak, P. M. Nayak and H. G. Joshi	Determinants Influencing the Entrepreneurial Success of MSMEs in Emerging Economies: A Study of Indian Women Entrepreneurs	2025	Cogent Economics & Finance	Studies factors affecting the success of MSMEs led by Indian women entrepreneurs.	Provides context-specific insights into success factors.	Questionnaire and sampling limitations; no automated recommendations.
13	M. Dhanalakshmi and E. Karthikeyan	Challenges and Future of Pradhan Mantri Mudra Yojana (PMMY) in Promoting Women Entrepreneurship in India	2025	Academic Research Journal of Science and Technology	Examines PMMY, financial inclusion and barriers for women entrepreneurs.	Highlights funding opportunities and barriers.	Focuses on a single scheme and has no technical implementation.
14	D. Salman, N. Fawzy and Z. Zaazou	Education in Empowering Egyptian Women Entrepreneurs	2025	Future Business Journal	Uses survey and PLS-SEM to study digitalisation and education.	Provides empirical findings on empowerment.	Egypt-focused; limited generalisation.
15	P. Prabha, A. Swati and R. Kumar	Enhancing Women Entrepreneurship Through Resource	2025	Women's Studies International	Uses SEM to study digital technologies and resource	Connects digitalisation with business	Lacks privacy and security mechanisms.

		Orchestration and Digital Technologies		Forum	orchestration in women's entrepreneurship.	growth.	
16	A. Pandhare, P. N. Bellampalli and N. Yadava	Transforming Rural Women's Lives in India: The Impact of Microfinance and Entrepreneurship on Empowerment in Self-Help Groups	2024	Journal of Innovation and Entrepreneurship	Studies microfinance and entrepreneurship in women's self-help groups.	Highlights financial inclusion and empowerment.	No digital privacy or security focus.
17	K. Shibano, K. Ito, H. Han, T. T. Chu, W. Ozaki, G. Mogi	Secure Processing and Distribution of Data Managed on Private InterPlanetary File System Using Zero-Knowledge Proofs	2024	Electronics	Uses private IPFS and zk-SNARKs for secure data verification.	Supports data integrity and privacy.	Limited large-scale evaluation.
18	B.-J. Chen, S. Waiwitlikhit, I. Stoica and D. Kang	ZKML: An Optimizing System for ML Inference in Zero-Knowledge Proofs	2024	Proceedings of EuroSys	Optimizes ML inference representation and proof generation using ZKPs.	Reduces ZKP proving cost.	ZKP remains computationally expensive.
19	B. Balaban et al.	Privacy-Preserving Machine Learning (PPML) Inference for Clinically Actionable Models	2025	IEEE Access	Applies privacy-preserving techniques to ML inference in healthcare.	Protects sensitive input data during inference.	Increased latency and computation.
20	L. Wu, X. A. Wang, J. Liu, Y. Su, Z. Tu, W. Liu, H. Lei, D. Tang, Y. Cao and J. Zhang	Homomorphic Encryption for Machine Learning Applications with CKKS Algorithms: A Survey of Developments and Applications	2025	Computers, Materials & Continua	Surveys CKKS-based HE applications in privacy-preserving ML.	Provides a detailed overview of CKKS applications.	Survey-based; no complete implementation.

3. METHODOLOGY OF EMPOWER-HER

Empower-Her is developed as a privacy-preserving AI framework designed to support women entrepreneurs in starting and developing their businesses. The system aims to provide business idea validation, government scheme recommendations, funding opportunities, market insights, and secure document verification through a unified web application.

To achieve these objectives, the system combines Artificial Intelligence (AI), Homomorphic Encryption (HE), Zero-Knowledge Proofs (ZKP), Optical Character Recognition (OCR), and web development technologies. These technologies work together to process user information, generate personalized recommendations, protect sensitive data, and present useful results through a centralized dashboard.

As illustrated in Figure 3.1, the methodology consists of six major stages: user input and registration, data processing and security, AI processing and analysis, document processing and verification, additional entrepreneurial services, and output generation through a personalized dashboard. The system also uses a knowledge base containing government schemes, market and industry data, financial information, business information, and safety resources. Continuous improvement is supported through user feedback and updated information.

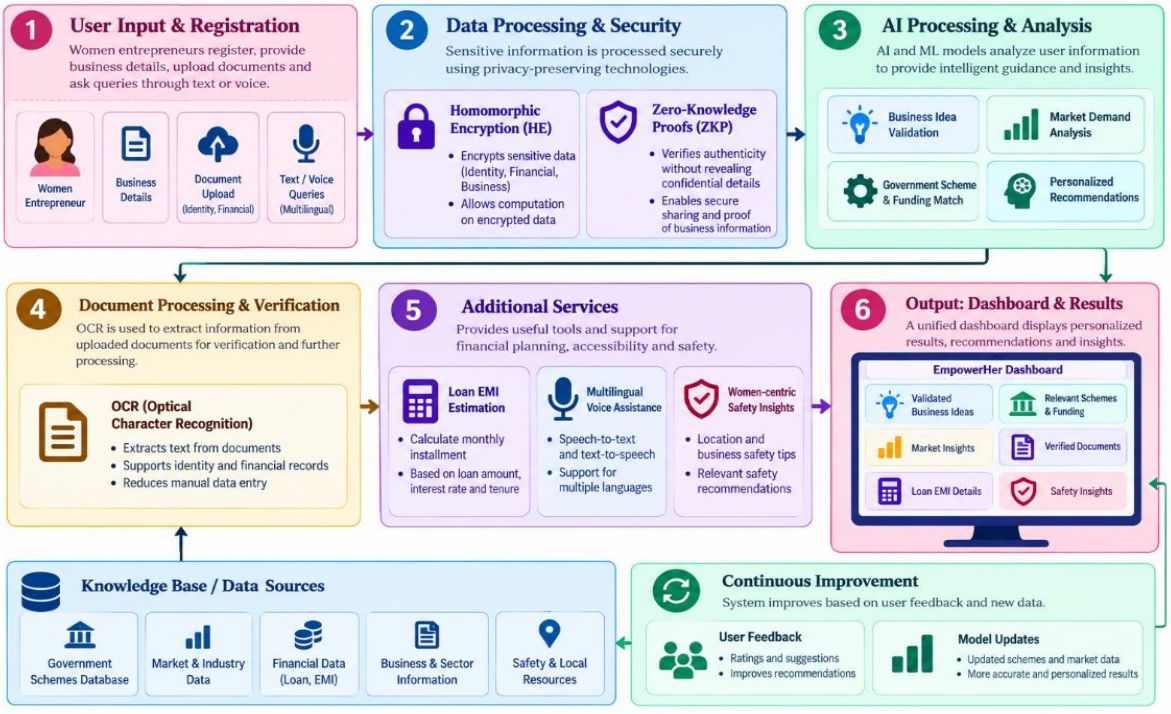


Figure 3.1: Proposed Methodology Flow of Empower-Her

3.1 Technologies Used:

The Empower-Her system uses a combination of programming languages, AI and machine learning libraries, cryptographic techniques, document-processing tools, and web technologies.

Python:

Python is used as the primary programming language for backend operations, data processing, AI integration, and recommendation generation. Its extensive library support makes it suitable for developing intelligent applications and handling business-related information.

Django and Django REST Framework

Django is used as the primary backend framework for developing the Empower-Her application. It manages the application's core business logic, user registration, authentication, request handling, and database interactions. Django REST Framework is used to develop REST APIs that enable communication between the frontend and backend. These APIs process user inputs, handle business idea submissions, and connect the frontend with AI-based analysis and other application modules. Django also supports access control and structured data management, helping maintain a secure and organized application.

Artificial Intelligence and Machine Learning

AI is used to analyze the business information submitted by users and generate useful suggestions. AI and machine learning techniques are used to analyze business ideas, understand user requirements, evaluate market demand, and generate recommendations. The proposed technology stack includes Scikit-learn, PyTorch, NLP libraries, and Large Language Models (LLMs), such as OpenAI models.

Privacy and Security Technologies

Homomorphic Encryption (HE) and Zero-Knowledge Proofs (ZKP) are used to support privacy-preserving computation and secure verification.

- Homomorphic Encryption: Enables supported computations to be performed on encrypted data without first decrypting it.
- Zero-Knowledge Proofs: Allow a user to prove that a statement is true without revealing the underlying confidential information.

Optical Character Recognition (OCR)

OCR is used to extract text from uploaded identity and financial documents. This helps reduce manual data entry and supports document verification and further processing.

The proposed tools include Tesseract OCR, EasyOCR, and OpenCV.

Recommendation System and Market Analysis

Python libraries such as Pandas, NumPy, and Scikit-learn support data processing, analysis, and recommendation generation. Market Data APIs and relevant business datasets can provide additional information for identifying potential customers, evaluating market demand, and generating business insights.

Voice Assistance

Voice assistance is intended to allow users to interact with the platform through speech and access information in multiple languages. The proposed technologies include SpeechRecognition, gTTS, pyttsx3, Whisper, and multilingual language models. These tools support speech recognition, speech generation, and voice-based interaction.

Backend and API Development

Django and Django REST Framework are proposed for backend development and API management. They handle user requests, authentication, business data processing, and communication between the frontend and backend components.

Database Management

PostgreSQL or SQLite is proposed for storing user details, business idea records, and other application information. The database supports organized data management and controlled access to stored records.

Frontend Development

React with Vite, HTML, CSS, JavaScript, and Bootstrap is proposed for developing the user interface. These technologies support interactive pages through which users can enter business information, upload documents, explore recommendations, and view their results.

3.2 Development Process

The development of Empower-Her follows a modular approach to provide AI-based business guidance and privacy-preserving support for women entrepreneurs. The proposed workflow consists of six major stages, as illustrated in Figure 3.1. These include user input and registration, data processing and security, AI-based analysis, document processing and verification, additional entrepreneurial services, and personalized dashboard generation.

The backend is developed using Python, Django, and Django REST Framework to manage user requests, authentication, APIs, and database operations. The business idea validation module is designed to analyze user-provided business details and generate

relevant recommendations. Additional modules are proposed for government scheme discovery, funding assistance, and market analysis.

The framework also proposes Homomorphic Encryption (HE) and Zero-Knowledge Proofs (ZKP) to support privacy-preserving computation and secure verification. OCR-based document processing, loan EMI estimation, multilingual voice assistance, and safety recommendations are included in the proposed design. These modules can be integrated progressively and tested independently.

Finally, the system is designed to present relevant information through a personalized dashboard. A knowledge base containing government scheme information, market data, financial resources, and business information supports the recommendation process. The modular architecture allows the application to be expanded as additional functionalities are implemented and tested.

4. RESULTS

This chapter presents the results of implementing and testing Empower-Her. The complete system described in Chapter 3 was built as a Django and Django REST Framework web application with ten modules. Each module was tested with automated tests and manually through the web interface. Section 4.1 summarises what was implemented, Section 4.2 describes the test setup, Sections 4.3 to 4.10 present the output of each module with screenshots, Section 4.11 reports performance measurements, Section 4.12 the testing results, and Section 4.13 discusses the results and their limitations.

4.1 Implementation Summary

Table 4.1 lists the modules of Empower-Her, the technique each one uses, and its status. All modules proposed in the methodology were implemented except the React frontend, which was replaced by server-rendered HTML pages that use the same REST APIs.

Table 4.1: Implemented modules of Empower-Her

Module	Technique / data used	Status
User registration & profile	Django auth, token authentication, PBKDF2 password hashing	Implemented
Business idea validation	Financial model, rule-based analysis, optional OpenAI LLM, confidential mode	Implemented
Government scheme recommendation	25 curated schemes; eligibility rules + TF-IDF similarity (scikit-learn)	Implemented
Funding & loan EMI	EMI, amortisation, affordability (FOIR), PMEGP subsidy	Implemented
Homomorphic encryption	Paillier (2048-bit) in browser and server	Implemented
Local market demand	Census 2011 (640 districts) + Ministry of MSME data	Implemented
Document verification (OCR)	RapidOCR, pypdf; PAN, Aadhaar (Verhoeff), GSTIN checksum, Udyam, IFSC	Implemented
Idea Vault with ZKP	Fernet encryption, Pedersen commitment, Okamoto/Schnorr proof	Implemented
Multilingual voice assistant	English, Telugu, Hindi; Web Speech API; optional LLM	Implemented
Women safety	Helplines, reporting portals, safety guides	Implemented

4.2 Experimental Setup

All results were obtained on a laptop with an Apple Silicon (arm64) processor running macOS (Darwin 25.5.0) and Python 3.12.10, with SQLite as the database. The application was started with the one-command setup script (setup.sh / setup.bat), which creates the virtual environment, generates encryption keys, prepares the datasets and database, and creates a demo account. The project requires Python 3.11 to 3.13 (the earlier requirement of Python 3.9 was raised by the data libraries used).

The demo account belongs to a fictitious entrepreneur, 'Priya Demo', from Warangal, Telangana, at the business-idea stage. It contains two validated ideas for millet snacks (one marked confidential), one idea in the Idea Vault, and a synthetic PAN card image. No real personal data was used. The datasets used are listed in Table 4.2.

Table 4.2: Datasets used

Dataset	Source	Size
District indicators	Census of India 2011, district tables	640 districts, 118 columns
Women entrepreneurship by state	Ministry of MSME annexures (PIB, Feb 2024): Udyam/UAP, CGTMSE, PMEGP	36 states/UTs
Government schemes	Curated from official scheme websites	25 schemes
Loan routes	Curated from scheme guidelines	9 loan products
Safety resources	Official helplines and portals	7 helplines, 4 portals, 6 guides

4.3 User Interface and Dashboard

Figure 4.1 shows the home page that visitors see before logging in. It introduces the platform and offers registration and login. After logging in, the entrepreneur reaches the dashboard (Figure 4.2), which gives one-click access to all nine modules. Every page shares a common navigation bar, and the layout adapts to mobile screens (Figure 4.19).

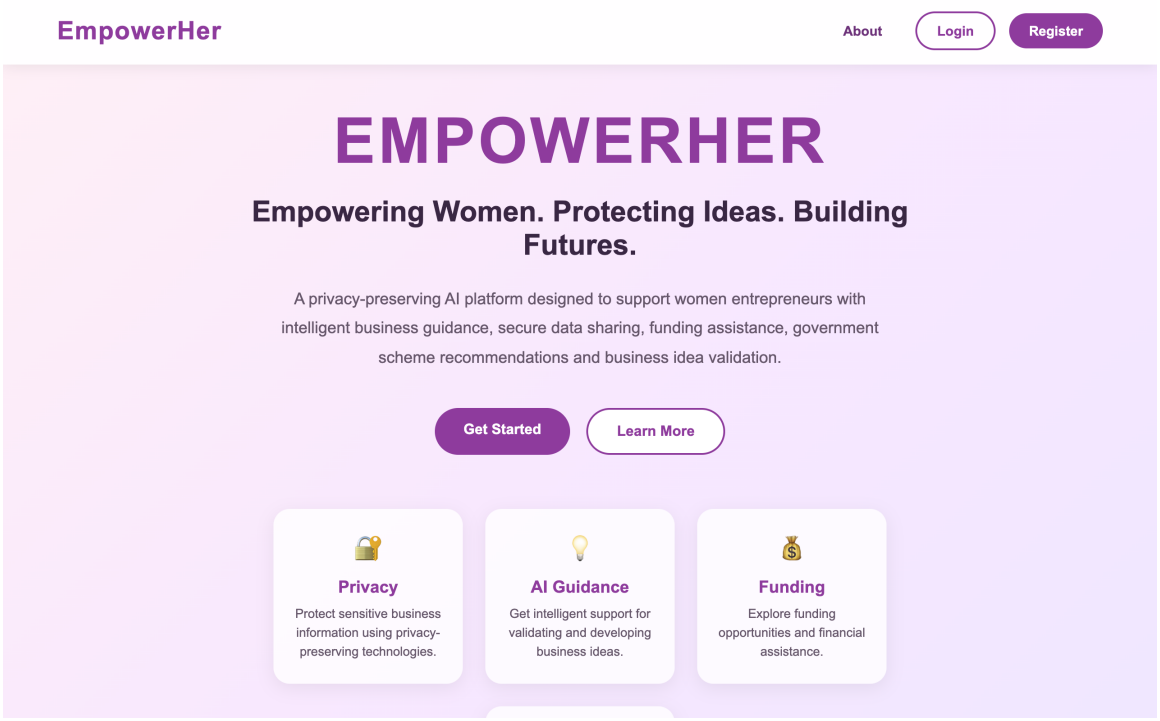


Figure 4.1: Home page of Empower-Her

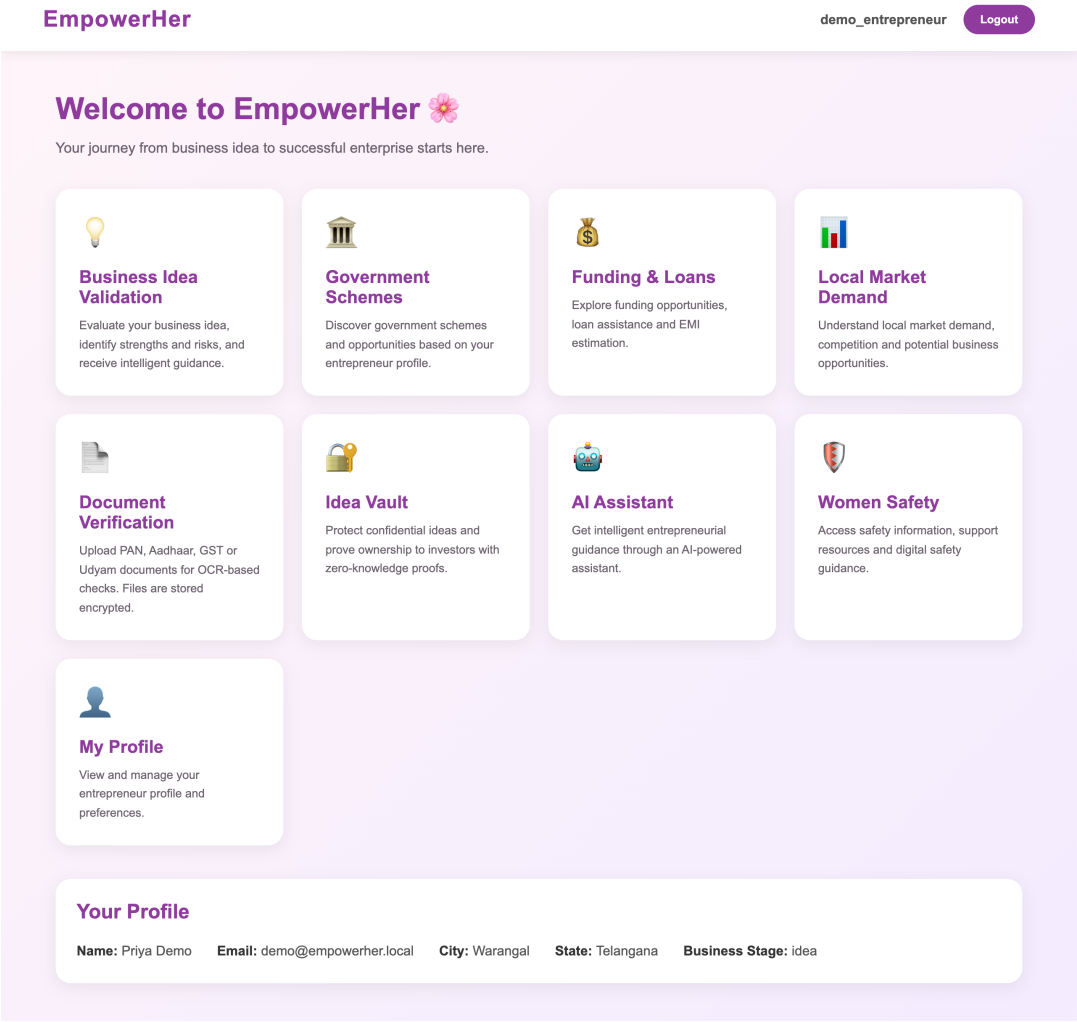


Figure 4.2: Entrepreneur dashboard with all modules

4.4 Business Idea Validation

The entrepreneur enters the business name, category, description, location, target customers and expected figures. Figure 4.3 shows the report for the 'Ragi Energy Bars' idea: with a selling price of Rs 120 and 600 units per month, the system calculates a monthly revenue of Rs 72,000, expenses of Rs 45,000, a profit of Rs 27,000 and a profit margin of 37.5%. The form completeness is 100%. The report then lists strengths, risks and practical suggestions.

This idea was submitted with the Confidential option (Figure 4.4). The note at the top of the report confirms that the description and target customers were stored encrypted and were not sent to the AI service, while the figures, category and location remained available for analysis. Inspection of the database confirmed that the description and target-customer columns were empty and only Fernet ciphertext was stored.

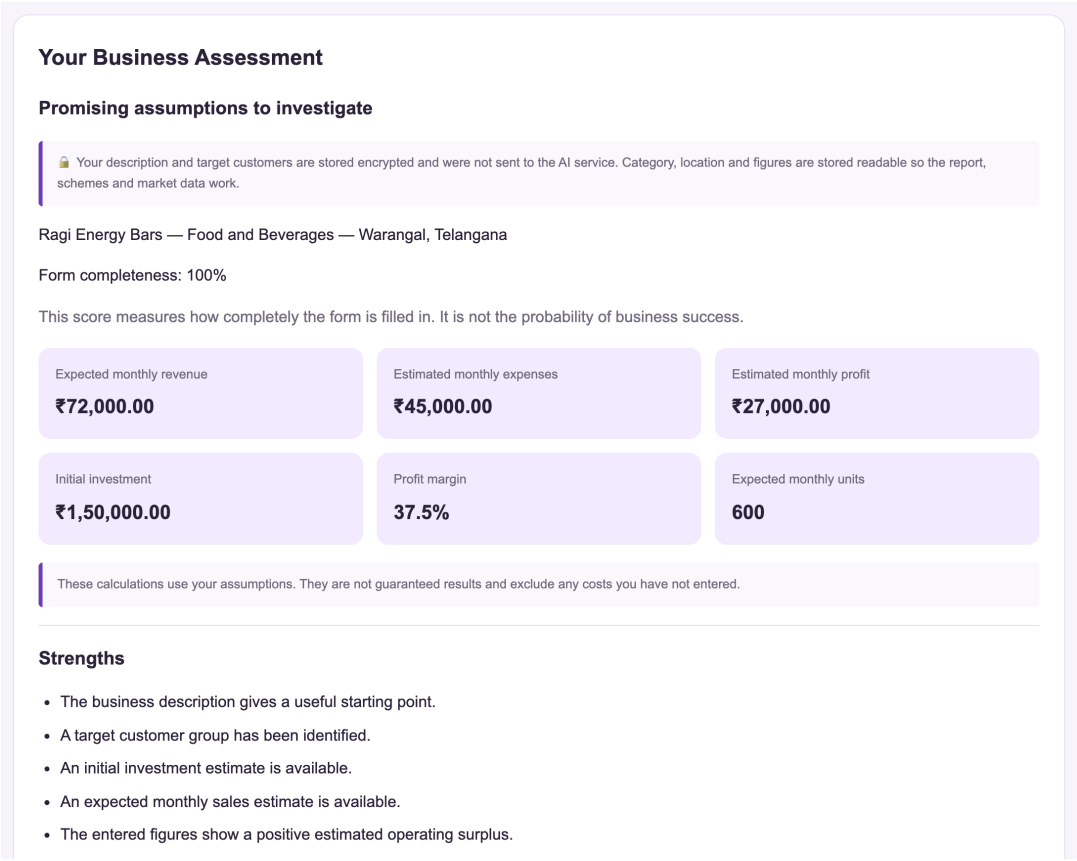


Figure 4.3: Business validation report for a confidential idea

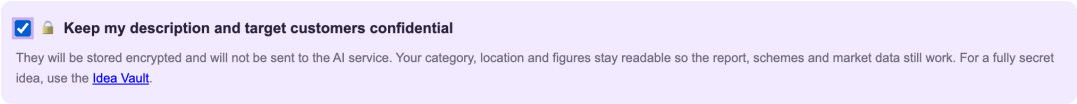


Figure 4.4: Confidential option on the validation form

Figure 4.5 shows the second half of the report. The market section places Warangal at a demand potential index of 70.5 out of 100 (Above average) for food businesses, ranking 4th of 10 Telangana districts. The report also recommends the three most relevant government schemes: WE Hub, Stree Nidhi and PM FME.

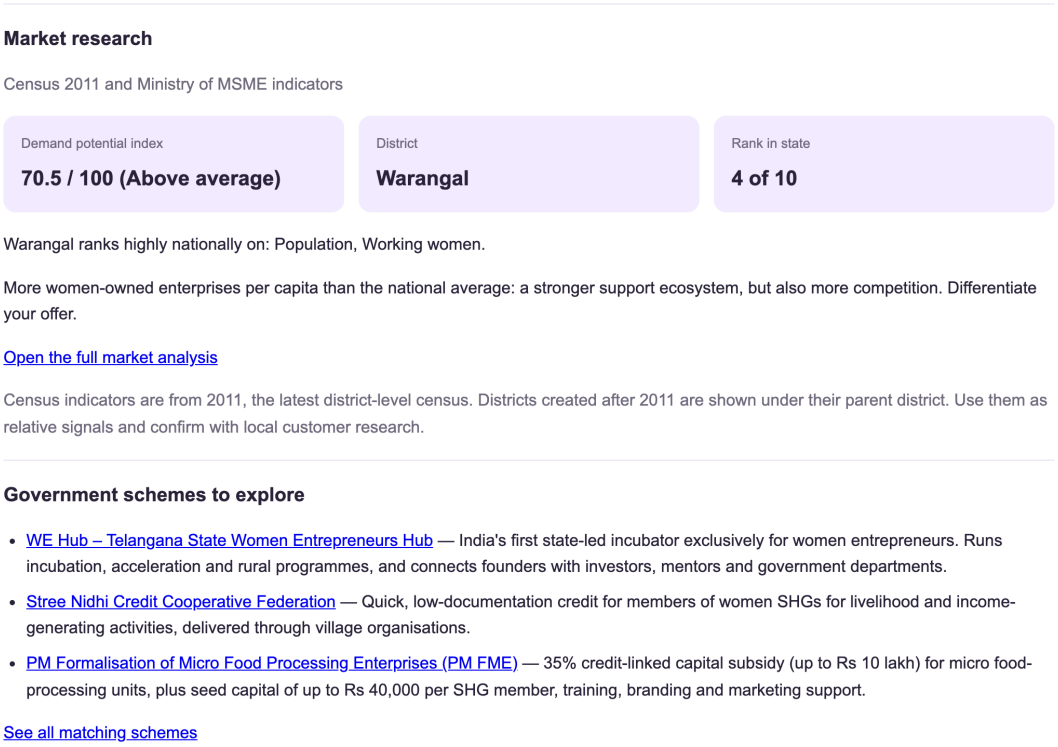


Figure 4.5: Market research and scheme recommendations inside the validation report

4.5 Government Scheme Recommendation

Figure 4.6 shows personalised recommendations for a food-business idea in Telangana at the idea stage, needing Rs 3,00,000, by a Self-Help Group member, described as 'homemade pickles and millet snacks'. The recommender first applies eligibility rules (state, business stage and social group), then scores the remaining schemes using category, amount and TF-IDF text similarity. The top results were PM FME (score 64.6), Stree Nidhi (63.0) and WE Hub (48.0). PM FME ranks first because it is designed for food processing, covers the amount needed, targets SHG members and matches words in the description. Each card lists these reasons, so the user can see why a scheme was suggested. Schemes that are reserved for groups the user does not belong to are excluded.

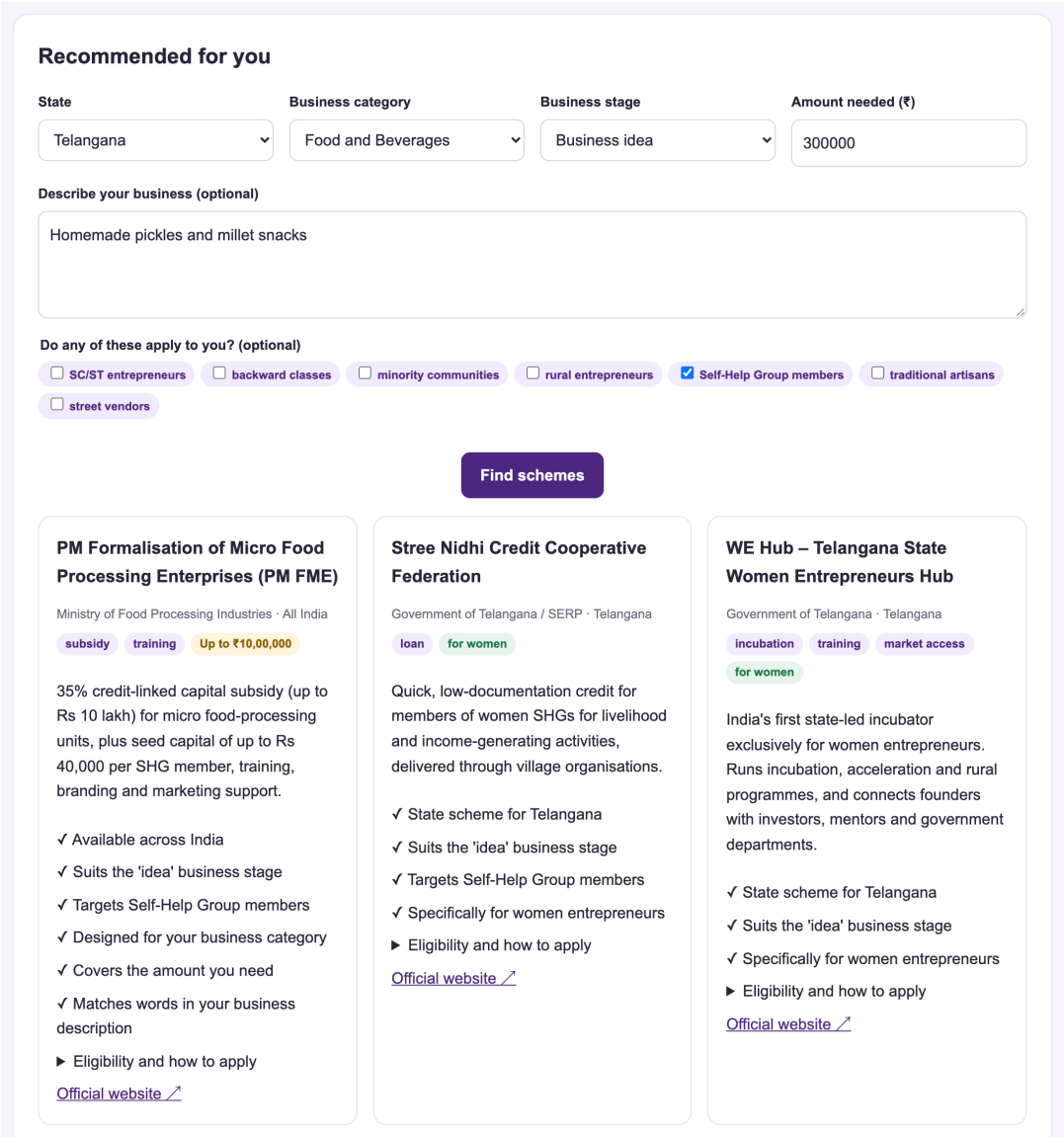


Figure 4.6: Personalised government scheme recommendations with reasons

4.6 Funding, EMI and Privacy-Preserving Affordability Check

Figure 4.7 shows the EMI calculator for a loan of Rs 5,00,000 at 10.5% per year for 60 months. The EMI is Rs 10,747 with a total interest of Rs 1,44,817. A year-by-year schedule and the loan routes that cover this amount are also shown. The EMI matches the standard reducing-balance formula; an independent check (Rs 5,00,000 at 10% for 36 months gives Rs 16,133.59) is part of the automated tests.

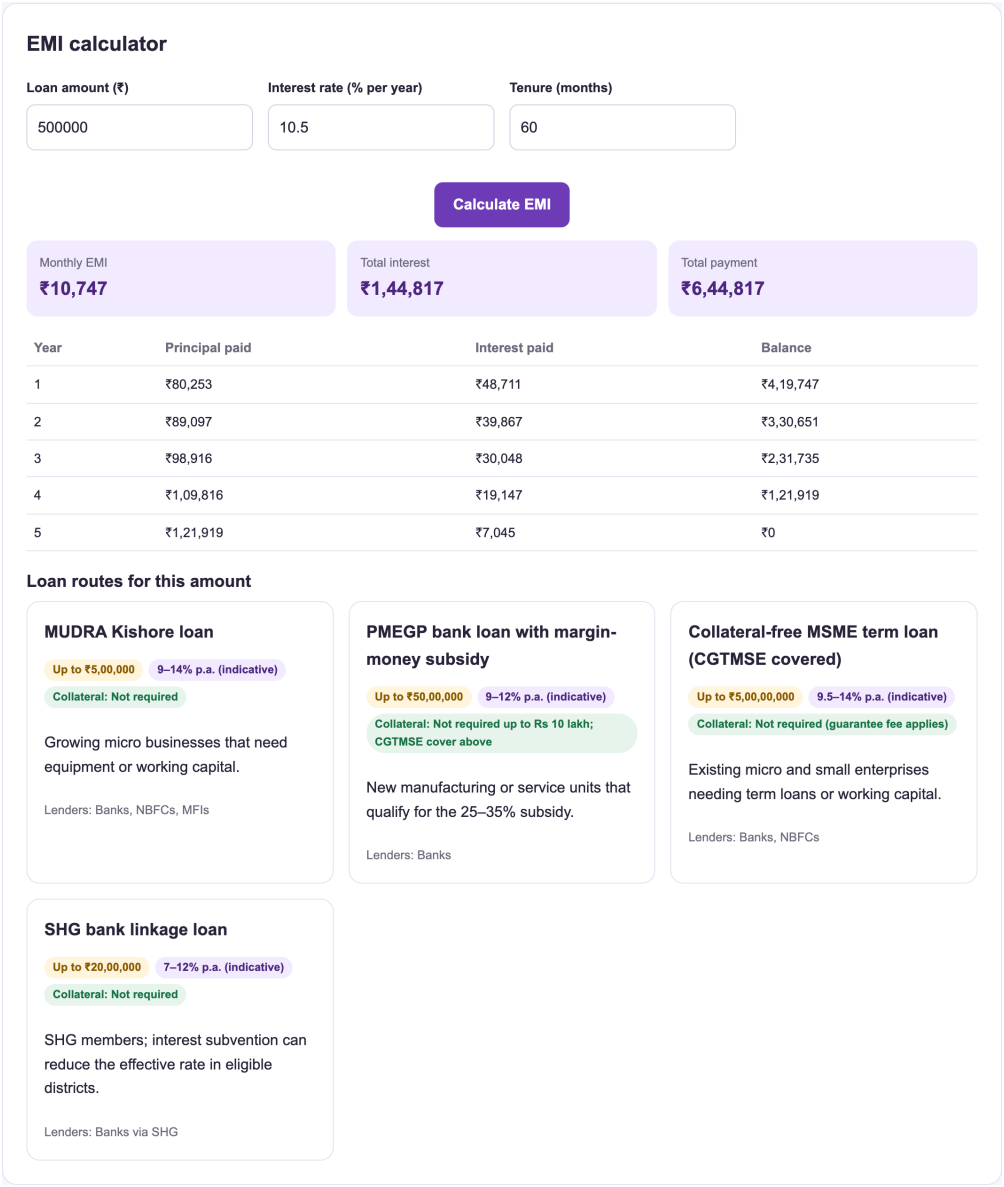


Figure 4.7: EMI calculator with repayment schedule

Figure 4.8 shows the privacy-preserving affordability check using Paillier homomorphic encryption. The browser generated a 2048-bit key pair, encrypted the monthly income (Rs 60,000), expenses (Rs 35,000) and existing EMIs (Rs 2,000), and sent only the public key and ciphertexts to the server. The 'What the server received' panel shows that the server saw only 4096-bit ciphertexts. The server computed the surplus and EMI headroom directly on the ciphertexts. After decryption in the browser, the results were a monthly surplus of Rs 25,000, a yearly surplus of Rs 3,00,000, total EMIs of Rs 11,000 and an EMI headroom of Rs 1,500. All four match the plaintext calculation, showing that the server can compute on financial data that it cannot read.

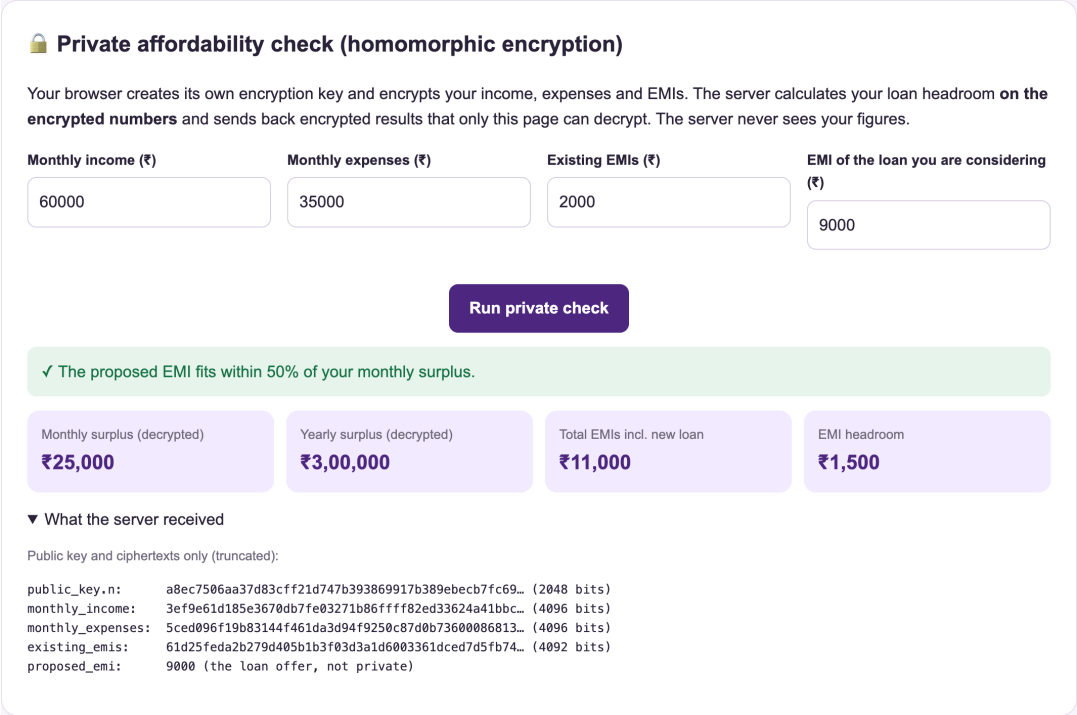


Figure 4.8: Private affordability check using homomorphic encryption

4.7 Local Market Demand Analysis

The market module compares a district with all 640 Indian districts using census indicators chosen for the business category. Figure 4.9 shows the result for food businesses in Warangal: a demand potential index of 70.5 (Above average), rank 4 of 10 in Telangana, with a population of 35,12,576 and 10,46,987 households (Census 2011).

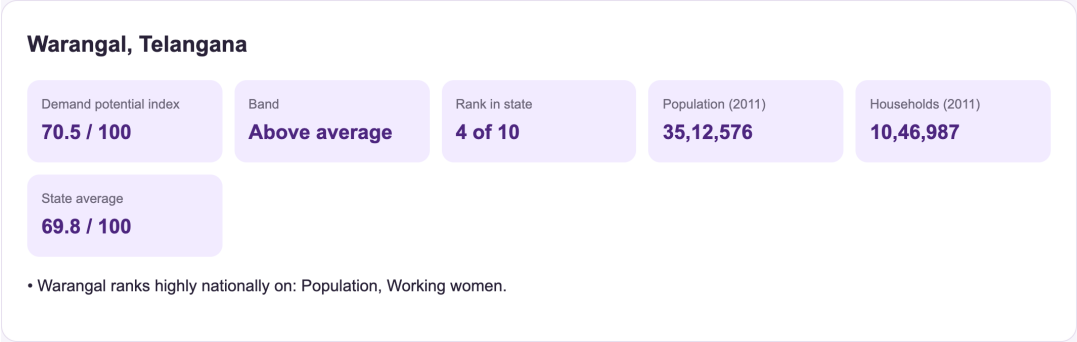


Figure 4.9: Market demand summary for Warangal (food and beverages)

Figure 4.10 explains the score. For food businesses the index weights population (30%), urban households (25%), working women, higher-income households and two-wheeler ownership (15% each). Warangal is in the 87th national percentile for population and for working women, but only the 42nd for higher-income households, which suggests that affordable pricing suits this market.

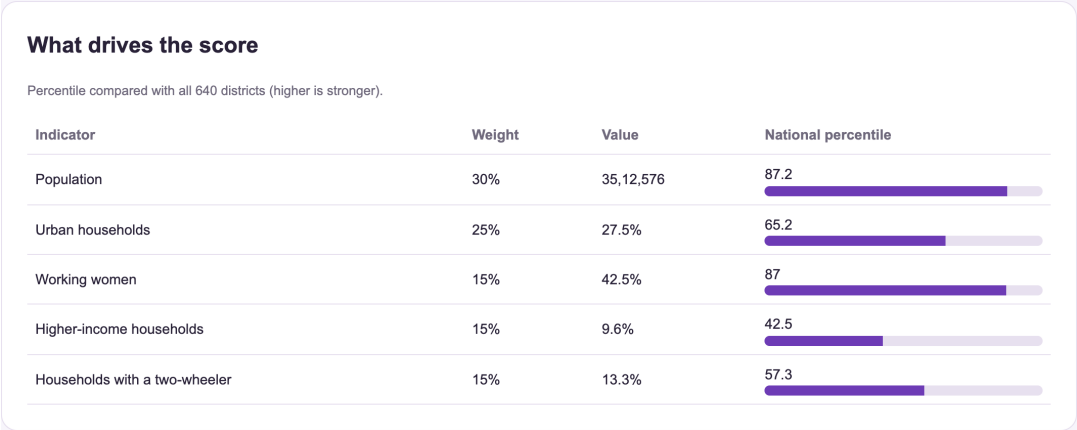


Figure 4.10: Indicators behind the demand potential index

Figure 4.11 shows the women-entrepreneurship ecosystem in Telangana from Ministry of MSME data: 5,79,210 women-owned MSMEs, or 33.12 per 1,000 women against a national average of 23.84, together with 45,803 CGTMSE guarantees and 5,963 PMEGP projects for women. The system interprets this as a stronger support ecosystem but also more competition.

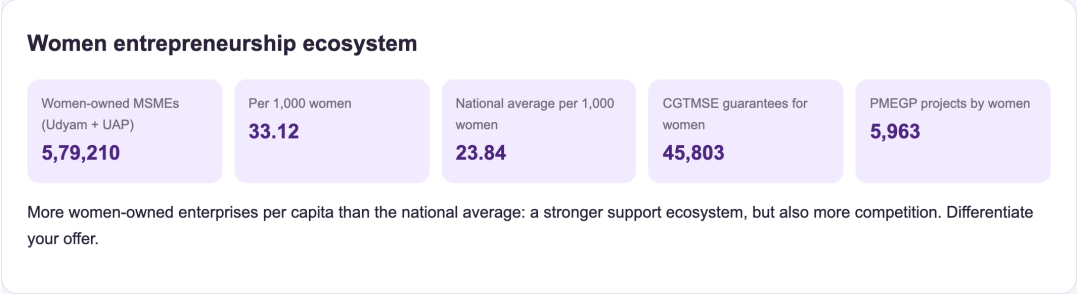


Figure 4.11: Women entrepreneurship ecosystem in Telangana

Figure 4.12 compares all business categories for Warangal. Retail (77.9), food (70.5) and local services (69.6) have the highest demand potential, while beauty and personal care (49.3) is just below the national middle. This helps an entrepreneur choose or position a business before investing.

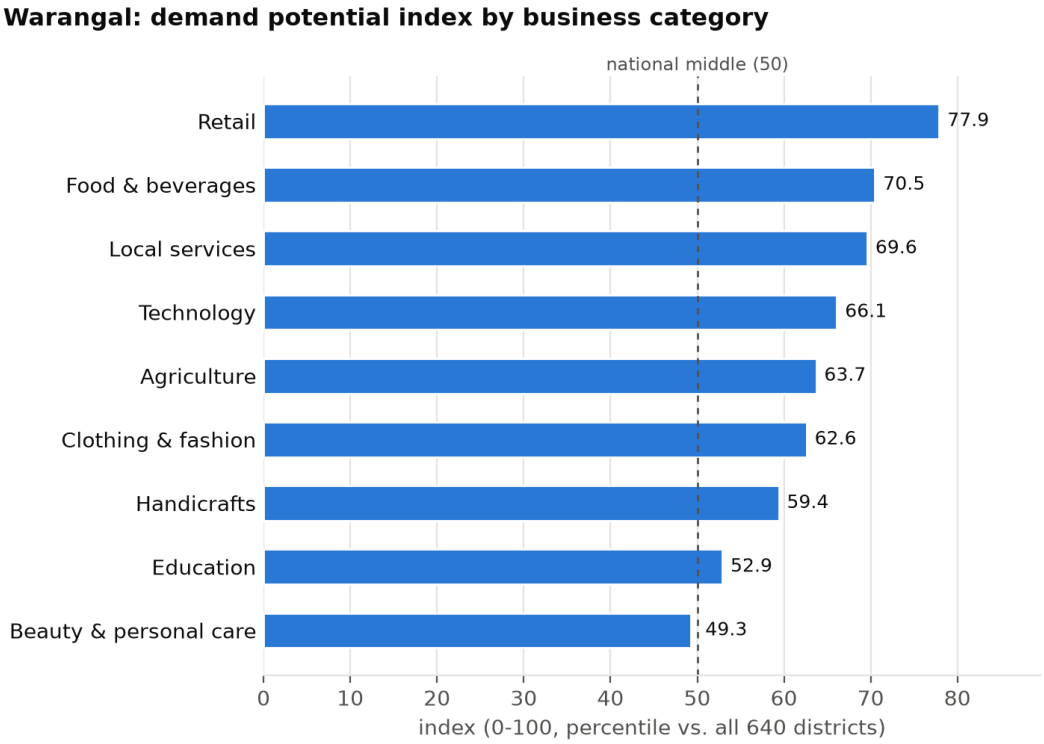


Figure 4.12: Demand potential index for each business category in Warangal

4.8 Document Verification using OCR

Figure 4.13 shows the result of uploading a synthetic PAN card image. The OCR engine read the card with 99.1% confidence, detected the document type as a PAN card automatically, validated the PAN structure, identified the holder type as 'Individual', and confirmed that the name on the card matches the user's profile. Only the masked number (ABXXX1234K) is stored; the original file and the full OCR text are encrypted. For Aadhaar cards the system validates the Verhoeff checksum and keeps only the last four digits, and for GSTINs it verifies the check character.

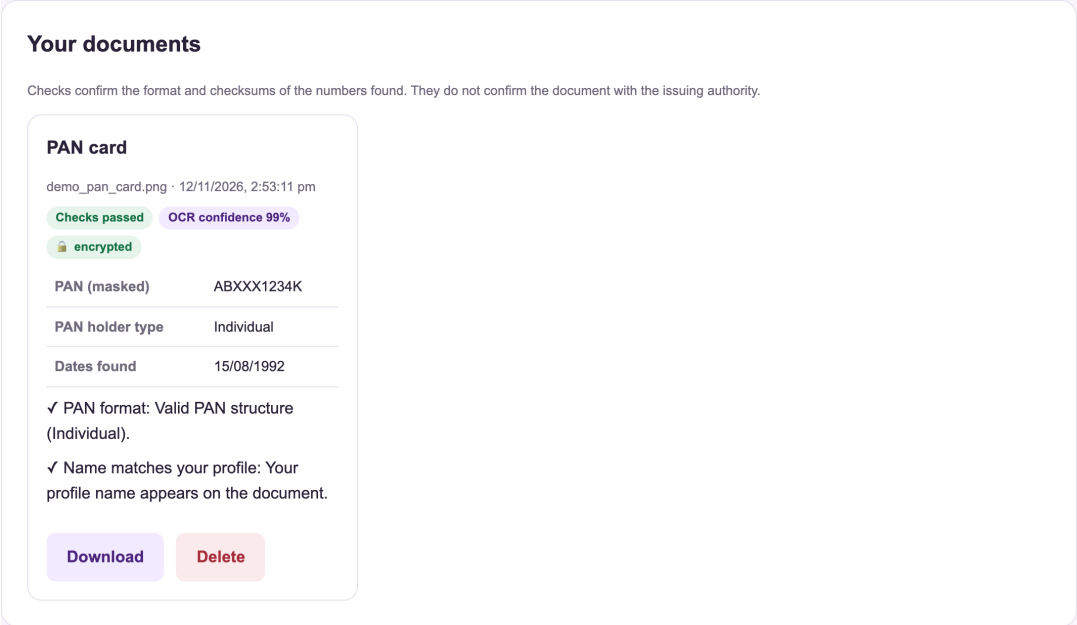


Figure 4.13: Document verification result for a PAN card

4.9 Idea Vault with Zero-Knowledge Proof

Figure 4.14 shows an idea registered in the Idea Vault. The idea text is encrypted, and a Pedersen commitment is stored with the registration time. Figure 4.15 shows the owner generating an ownership proof for a specific recipient ('Shared with WE Hub mentors for pitch review, Nov 2026') using the witness secret that only the owner holds.

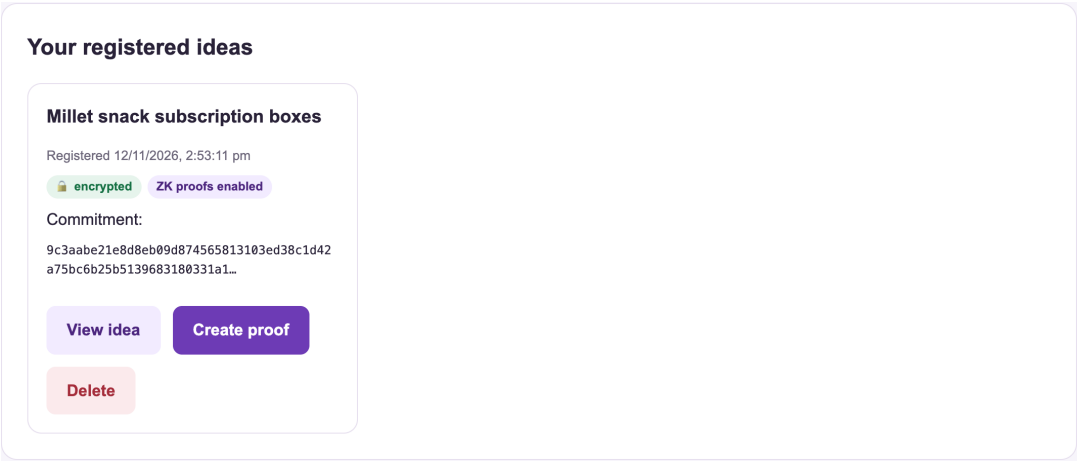


Figure 4.14: Idea registered in the Idea Vault

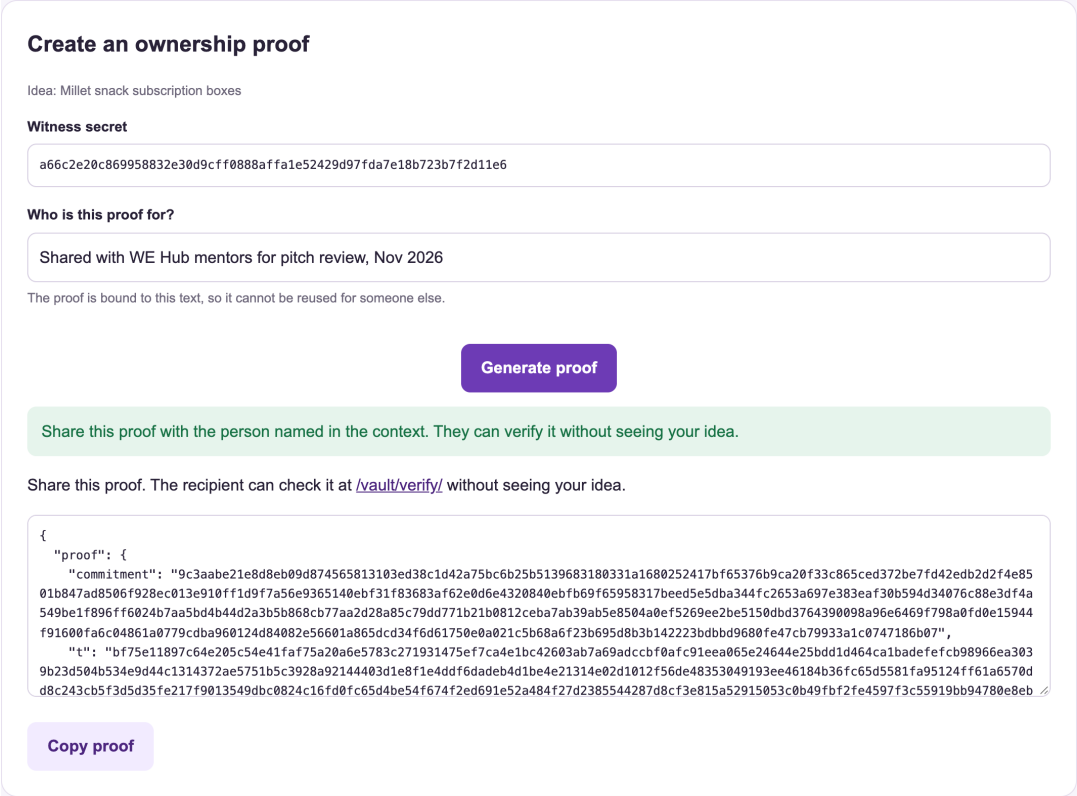


Figure 4.15: Generating a zero-knowledge ownership proof

Figure 4.16 shows the public verification page, which an investor or mentor can use without an account. The proof was accepted: it is valid, the commitment is registered on Empower-Her, and the registration time and recipient are shown. The verifier never sees the idea itself. In testing, proofs that were altered, or reused with a different recipient text, were rejected.



Figure 4.16: Successful verification of an ownership proof

4.10 Multilingual Assistant and Safety

Figure 4.17 shows the assistant answering in Telugu and Hindi. Asked in Telugu for the EMI of a Rs 5 lakh loan at 10% for 3 years, it replied with an EMI of Rs 16,134 and a total interest of Rs 80,809. Asked in Hindi which government schemes suit the user, it listed schemes matching her Telangana profile with official links. Questions can also be spoken through the microphone button, and answers read aloud. Without an AI key the assistant answers only from Empower-Her's verified data, which avoids invented schemes or numbers.

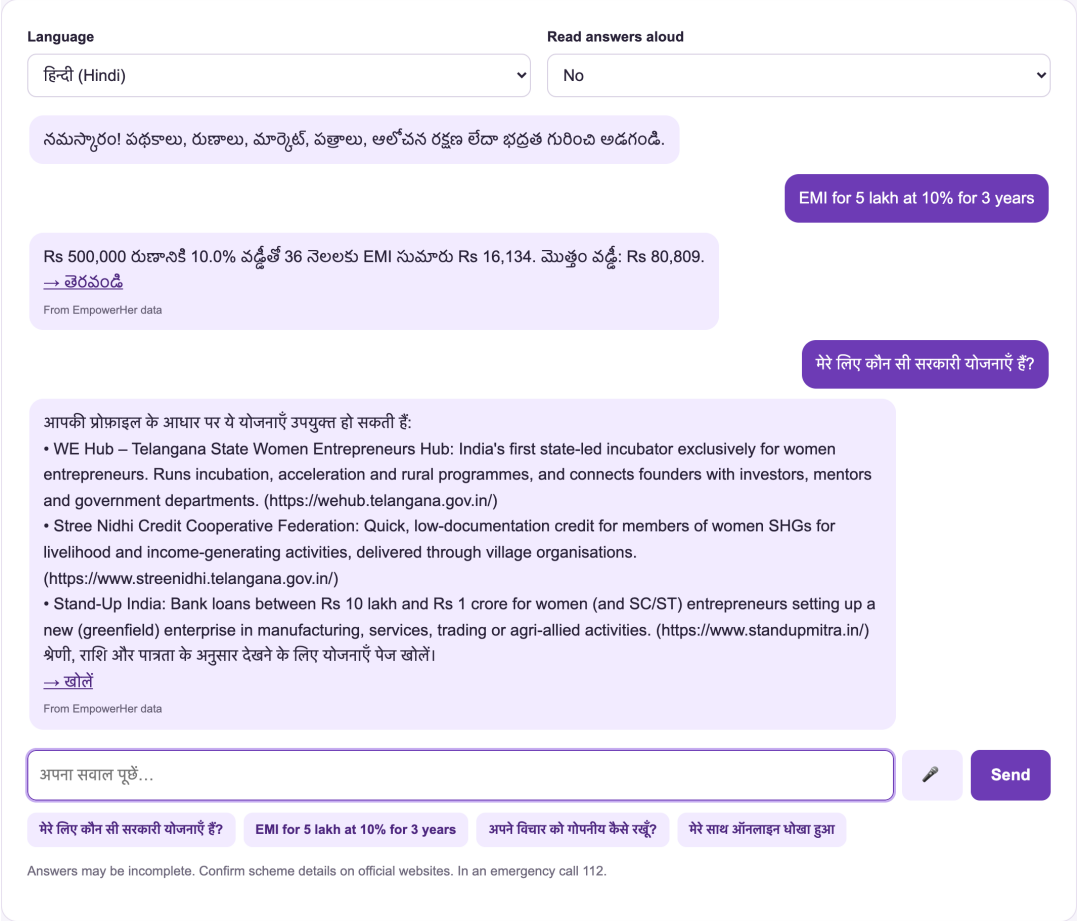


Figure 4.17: Assistant answering in Telugu and Hindi

Figure 4.18 shows the Women Safety module with a prominent emergency call button (112), national helplines and the Hyderabad SHE Teams number for Telangana users, reporting portals and practical safety guides. Figure 4.19 shows the funding page on a mobile phone, demonstrating the responsive layout.

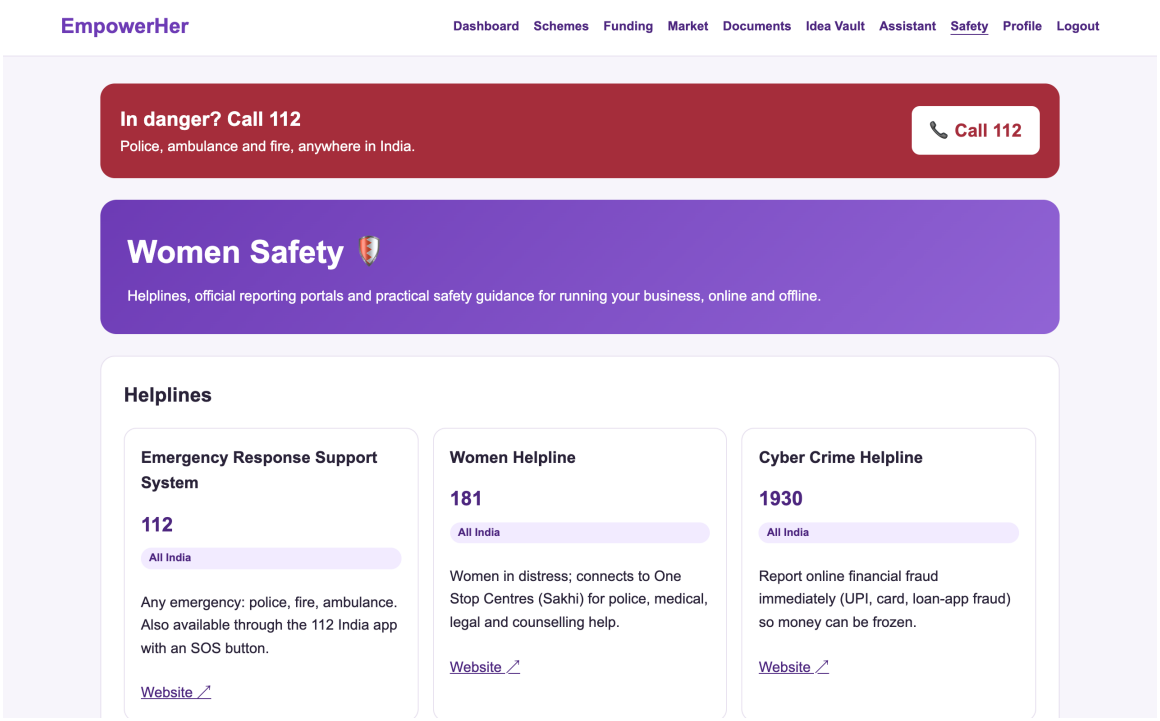


Figure 4.18: Women Safety module

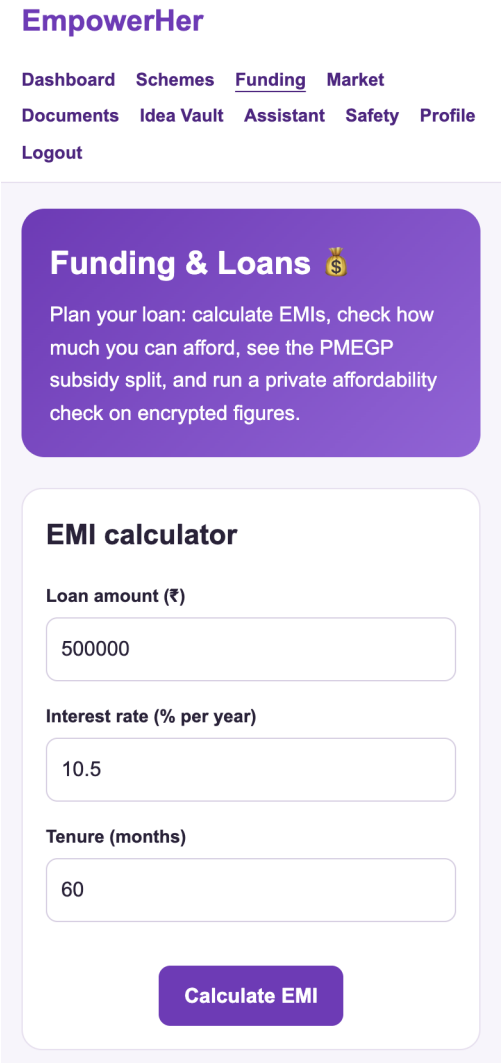


Figure 4.19: Funding page on a mobile phone

4.11 Performance Results

The performance of each operation was measured with the run_benchmarks command; Table 4.3 and Figure 4.20 report the median time over repeated runs. Everyday operations such as Fernet encryption, EMI calculation, scheme recommendation and market analysis complete in under 5 milliseconds. The cryptographic operations (Paillier and ZKP) take between 30 and 500 milliseconds because they use 2048-bit numbers. This is acceptable for interactive use: the complete private affordability check, including key generation in the browser, finished in about 1.2 seconds.

Table 4.3: Measured performance (median of repeated runs)

Operation	Median time	Notes
Fernet encryption of a 1 KB idea	0.006 ms	1000 B -> 1420 B ciphertext
Fernet decryption	0.005 ms	Authenticated (HMAC-SHA256)

Operation	Median time	Notes
Paillier key generation (2048-bit)	494.0 ms	Done once per check
Paillier encryption of one value	65.6 ms	4096-bit ciphertext
Affordability computed on ciphertexts	68.3 ms	Decrypted result 150,000 = expected 150,000
Paillier decryption of one value	65.7 ms	
ZKP commitment	31.2 ms	2048-bit group (RFC 3526)
ZKP proof generation	62.7 ms	Proof size 2,128 bytes
ZKP proof verification	82.7 ms	Tampered and replayed proofs rejected
OCR of a PAN card image	232.3 ms	99.1% confidence
Scheme recommendation	0.327 ms	25 schemes
Market analysis	4.6 ms	640 districts
EMI and schedule	0.008 ms	

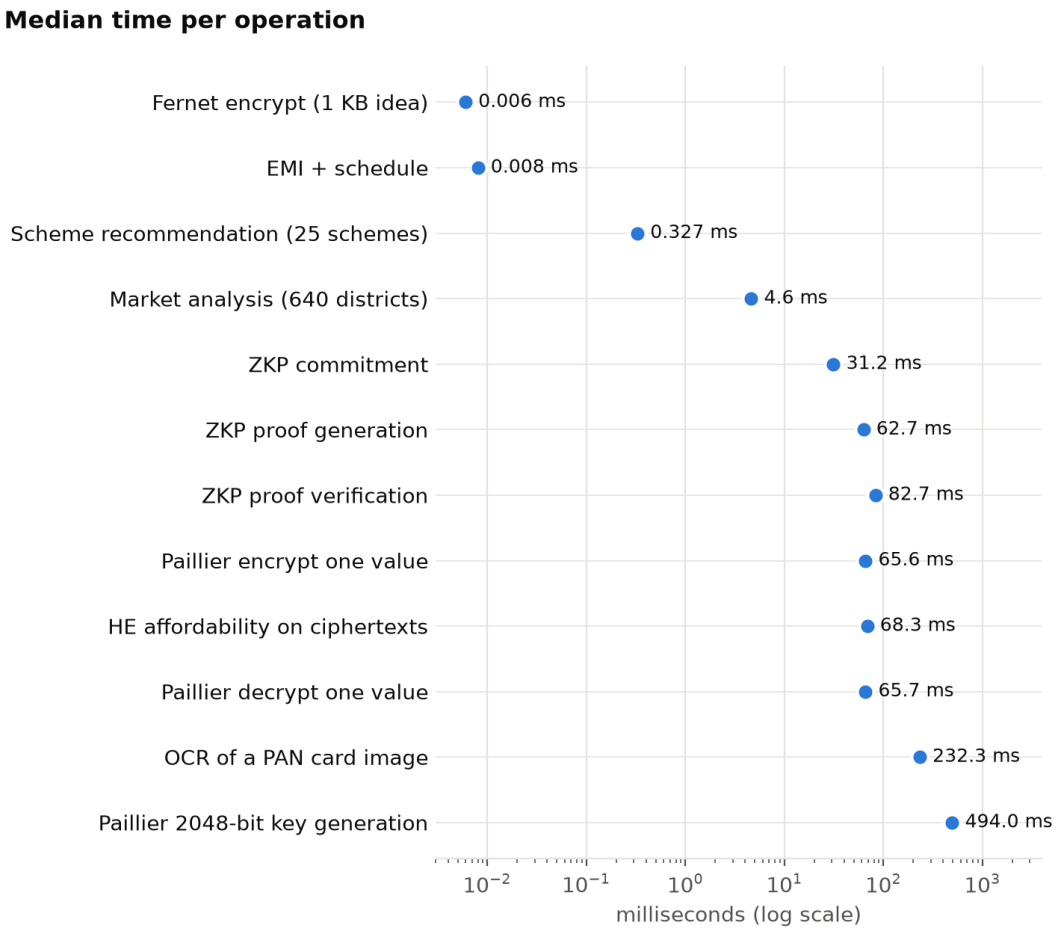


Figure 4.20: Median time per operation (logarithmic scale)

4.12 Testing Results

The system has 49 automated tests covering every module, and all 49 pass (Figure 4.21). The tests check correctness, such as EMI values, Paillier homomorphic results, ZKP verification, OCR field extraction and Aadhaar and GSTIN checksums. They also check security: owner-only access to ideas and documents, rejection of tampered proofs and fake file types, masking of identity numbers, and that confidential ideas are never sent to the AI service. In addition, the setup script was tested on a fresh copy of the project: it installed everything, rebuilt the datasets from the raw downloads, ran all tests and started the server in about one minute.

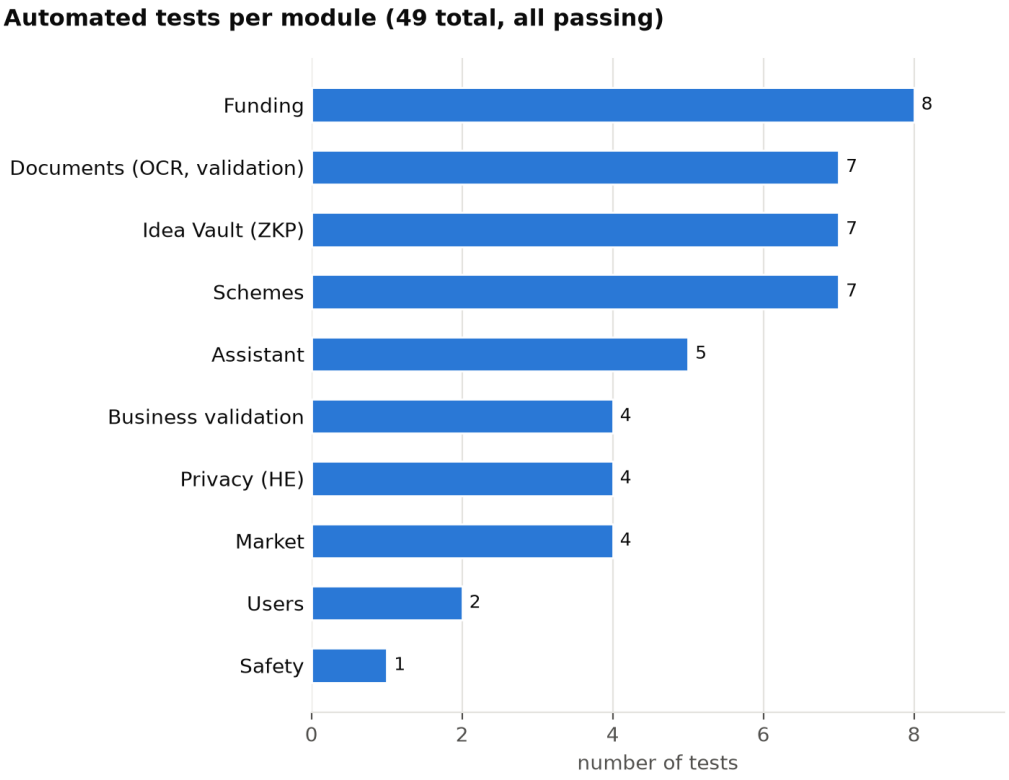


Figure 4.21: Automated tests per module

4.13 Discussion

The results show that Empower-Her meets the objectives stated in Chapter 1. It provides a single web application for business-idea validation, scheme and funding discovery, market insight and personalised guidance. It also shows that privacy-preserving techniques can be used in practice: homomorphic encryption allowed loan affordability to be computed without revealing income, and zero-knowledge proofs allowed ownership of an idea to be proven without disclosing it, both within interactive response times.

The results also have limitations:

- The market indicators come from Census 2011, the latest district-level census, so they show relative demand between districts, not current market size.
- Scheme details change over time and must be confirmed on the official websites linked from each scheme.
- Document checks confirm the format and checksums of numbers, not the authenticity of the document with the issuing authority.
- The server holds the encryption keys for stored ideas and documents. This protects against database leaks and against verifiers, but not against the server operator.
- Performance was measured on a single laptop with synthetic data, not under multi-user load.

4.CONCLUSION AND FUTURE SCOPE

4.1 CONCLUSION

Empower-Her is proposed as a privacy-preserving AI framework designed to support women entrepreneurs in developing their business ideas and accessing relevant entrepreneurial resources. The project aims to address challenges such as limited access to business guidance, difficulty in identifying suitable government schemes and funding opportunities, and concerns regarding the privacy of sensitive business information.

The proposed system combines Artificial Intelligence, web technologies, and privacy-preserving techniques to provide business idea validation, market insights, personalized recommendations, and secure information handling. Homomorphic Encryption (HE) and Zero-Knowledge Proofs (ZKP) are proposed to strengthen data privacy and support secure verification. Additional features, such as OCR-based document processing, loan EMI estimation, and multilingual voice assistance, are included in the overall system design.

The modular architecture of Empower-Her provides a foundation for integrating these functionalities into a unified platform. Although some modules are still under development, the proposed framework establishes a direction for building an accessible and privacy-focused entrepreneurial support system for women in both rural and urban areas.

4.2 FUTURE SCOPE

Empower-Her can be enhanced in the future by adding multilingual voice assistance and advanced AI models to provide better business guidance. Integration with updated government schemes, funding opportunities, and market datasets can improve recommendation accuracy.

The system can also be extended with Homomorphic Encryption, Zero-Knowledge Proofs, and automated document processing to strengthen privacy and security. Additional features such as financial planning and local market analysis can make the platform more useful for women entrepreneurs in rural and urban areas.

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